

PERSONAL DATA AND CONTACT INFORMATION

Name: Michal Kasarda
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EDUCATION

- **Faculty of Electrical Engineering, CTU in Prague** *since 2023, graduated with honors 2025*
master's degree in Robotics and Cybernetics
- **Faculty of Electrical Engineering, CTU in Prague** *since 2019, graduated 2023*
bachelor's degree in Robotics and Cybernetics
- **Gymnázium Stropkov, Slovak Republic** *since 2014, graduated 2018*
graduated in general education

EXPERIENCE

- **CTU in Prague** *June 2024 – Present*
Phd student, Vision for Robotics and Autonomous Systems (VRAS) Group *Prague*
 - Kolibriq Project – Research on deep learning methods to aid UAV command and control in cooperation with Lockheed Martin. My work focuses mainly on wind estimation in close proximity to the terrain for UAV planning purposes.
- **Czech Institute of Informatics, Robotics and Cybernetics** *March 2023 – September 2024*
Undergraduate Researcher, Intelligent Mobile Robotics (IMR) Group *Prague*
 - Research on visual teach-and-repeat navigation using visual place recognition based on deep-learned local features.
 - Developed courses on the ROS2 MoveIt framework, published on the Skillsmove portal.

PUBLICATIONS

- **Volumetric Wind Prediction Around Obstacles from Onboard UAV Measurements**
diploma Thesis — defended June 2025, received Dean's Award
 - Developed a real-time neural approximation of CFD solvers for high-resolution urban wind field prediction.
 - Implemented a fully differentiable UAV dynamics and control model for planning and optimization in predicted wind fields.
 - **Diploma thesis (full text)**
- **Multi-Platform Teach-and-Repeat Navigation by Visual Place Recognition Based on Deep-Learned Local Features**
conference paper, in review, submitted August 2025
 - I co-developed an improved method for teach-and-repeat navigation by introducing a novel approach to compute lateral and rotational offsets between the current image captured from the robot's pose and the query image.
 - **arXiv:2503.13090**

SKILLS AND INTERESTS

Programming Languages: C, C++, Python
Developer Tools: Jupyter, Visual Studio Code, shell (bash), OpenFoam, Ansys Fluent
Technical Areas of Interest: deep (reinforcement) learning, localization and mapping in robotics, computational fluid dynamics
Languages: Slovak (native), English B2